The Canadian Journal of Statistics
Vol. 32, No. 1, 2004, Pages 27–37
La revue canadienne de statistique

Nonlinear mixed-effect models with nonignorably missing covariates

Lang WU

Key words and phrases: EM algorithm; Gibbs sampling; longitudinal data; missing data; rejection sampling.

MSC 2000: Primary 62J02; secondary 62P10.

Abstract: Nonlinear mixed-effect models are often used in the analysis of longitudinal data. However, it sometimes happens that missing values for some of the model covariates are not purely random. Motivated by an application to HIV viral dynamics, where this situation occurs, the author considers likelihood inference for this type of problem. His approach involves a Monte Carlo EM algorithm, along with a Gibbs sampler and rejection/importance sampling methods. A concrete application is provided.

Modèles à effets mixtes non linéaires avec covariables manquantes et pour cause

Résumé : Les modèles à effets mixtes non linéaires sont souvent employés pour l'analyse de données longitudinales. Il arrive toutefois que les valeurs manquantes de certaines covariables du modèle ne soient pas le fruit d'un pur hasard. Motivé par des travaux sur la propagation du VIH, où cette situation se produit, l'auteur s'intéresse à l'inférence fondée sur la vraisemblance pour ce type de problème. Son approche s'appuie sur un algorithme EM de Monte-Carlo, un échantillonneur de Gibbs et des méthodes d'échantillonnage pondéré à critère de rejet. Une application concrète est présentée.

1. INTRODUCTION

For longitudinal data or repeated measurement data, nonlinear modelling is often required for a meaningful analysis in many biological applications, such as studies of HIV viral dynamics, pharmacokinetic analyses, and studies of growth and decay. Nonlinear mixed-effect models or hierarchical nonlinear models are powerful tools for such studies in characterizing the intra-individual variation and the interindividual variation (Davidian & Giltinan 1995; Vonesh & Chinchilli 1997). Understanding the nature of the interindividual systematic and random variation often receives great emphasis. Much of the interindividual variation may be explained by baseline covariates such as demographic and physiological information. In practice, however, many of these covariates may contain missing data. In some cases, the missing covariates should not be ignored since the state of missingness might be related essentially to the values that are missing (Little & Rubin 1987). In this article, we investigate problems of missing covariates in nonlinear mixed-effect models when the missing data mechanism is nonignorable.

Our research is motivated by studies of HIV viral dynamics. In recent years, HIV viral dynamic models have received great attention in AIDS studies (Ho et al. 1995; Perelson et al. 1996; Wu & Ding 1999). These viral dynamic models provide a good understanding of the pathogenesis of HIV infection and evaluation of antiretroviral therapies. The dynamic parameters may reflect the efficacy of the antiviral treatments. Nonlinear mixed-effect models have been proposed in these studies for modelling the intra- and interpatient variations (Wu & Ding 1999). In these contexts, the interpatient variation is often large, and understanding this interpatient variation has received a great deal of attention since it may help us to make better clinical decisions and provide individualized treatments. Previous studies show that the interpatient variation may be partially explained by baseline covariates. Statistical analyses are complicated by missing values in the covariates. Wu & Wu (2001) proposed a simple multiple imputation method for missing covariates in nonlinear mixed-effect models under the ignorable assumption (i.e., the missing data are missing at random, Little & Rubin 1987). However, this missing-at-random as-

sumption may be difficult to verify. Moreover, in many AIDS studies, covariates may be missing for essential factors which cannot be ignored and therefore, the fact that they are missing would be nonignorable.

When the missing data mechanism is nonignorable, serious biases in the parameter estimates may occur if we do not model the missing data mechanism (Little & Rubin 1987). It is well known that the commonly used complete case method, which discards all incomplete observations, may give misleading results. There is an extensive literature on missing covariates in regression models (e.g., Chen & Little 1999; Ibrahim, Lipsitz & Chen 1999). Little (1992) gave an excellent survey. Ibrahim, Lipsitz & Chen (1999) considered missing covariates in generalized linear models and Ibrahim, Chen & Lipsitz (2001) investigated missing responses in generalized linear mixed models when the missing data mechanism is nonignorable. There is, however, little literature on missing covariates in nonlinear mixed-effect models.

In this article, we consider Monte Carlo EM algorithms for likelihood inference for nonlinear mixed-effect models with nonignorably missing covariates. We allow for an arbitrary number of missing covariates and arbitrary missing data patterns. The covariates can be categorical, continuous, or mixed. Our method is similar to that of Ibrahim, Lipsitz & Chen (1999) and Ibrahim, Chen & Lipsitz (2001), but we focus on nonlinear mixed-effect models rather than the generalized linear or the generalized linear mixed models and explore different sampling methods for the E-step. We extend the method of Ibrahim, Lipsitz & Chen (1999) to nonlinear mixed-effect models with nonignorably missing covariates.

The article is organized as follows. Section 2 presents an example which motivates our research. Section 3 describes the general models, the method, and various sampling strategies. The data analysis is presented in Section 4. We conclude the article with some discussion in Section 5.

2. AN EXAMPLE

We consider an AIDS study which motivates our research. In this study, 53 HIV-1 infected patients are given a potent antiretroviral treatment. Five patients discontinued the study due to drug intolerance and other problems, and these five patients are excluded in the analyses. The viral load (plasma HIV-1 RNA) is repeatedly measured over time for a period of about 48 weeks. As in Wu & Ding (1999), we consider only the viral load data before rebound, and we also remove data after three months, since these data are likely to be contaminated by long-term clinical factors. For the remaining data, the number of viral load measurements for each patient varies from 2 to 8, mostly due to different schedules or viral rebound.

The data exhibit large interpatient variations. Previous studies show that these interpatient variations may be partially explained by baseline covariates such as baseline CD4 cell counts, tumor necrosis factor (measured by plasma TNF levels), and total complement levels (measured by CH50). It is important to understand the mechanisms of the inter-patient variation in order to make clinical decisions and provide individualized treatments. The following two-phase HIV viral dynamic model has been proposed for this study (Wu & Ding 1999; Wu & Wu 2001)

$$y_{ij} = \log_{10}(P_{1i}e^{-\lambda_{1i}t_{ij}} + P_{2i}e^{-\lambda_{2i}t_{ij}}) + e_{ij}, \quad (1)$$

$$\log(P_{1i}) = \log(P_{10}) + a_{11}TNF + b_{1i}, \quad \lambda_{1i} = \lambda_{10} + a_{21}TNF + a_{22}CD4 + b_{2i}, \quad (2)$$

$$\log(P_{2i}) = \log(P_{20}) + a_{31}TNF + b_{3i}, \quad \lambda_{2i} = \lambda_{20} + a_{41}CH50 + b_{4i},$$

where y_{ij} is the $\log_{10}$ transformation of the viral load measurement for the ith patient at the jth time point, λ_{1i} and λ_{2i} represent two viral decay rates, and P_{1i} and P_{2i} are baseline values.

In this study, the covariates CD4 (z_1), CH50 (z_2), and TNF (z_3) contain 20.5%, 19.0%, and 16.7% missing data, respectively, so standard complete-data methods are not directly applicable. In previous studies (Wu & Ding 1999; Wu & Wu 2001), the missing covariates are assumed to be ignorable (missing at random) and simple imputation methods were used to handle the missing

1708945x, 2004, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.2307/3315997 by Northeast Normal University, Wiley Online Library on [26/03/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

covariates. Although the missing-at-random assumption may be reasonable in many studies, it is *not verifiable* based on the observed data. In AIDS studies, it is common that many missing data may be nonignorable, in that the missingness may be related essentially or circumstantially to the values that are missing. For example, it is conceivable that a patient with very low (or high) CD4 values may miss measurements because he/she is too sick (or healthy). However, it may not always be easy to find out the reasons for each missing value. When the missing data are nonignorable, statistical analyses assuming these to be missing at random may give severely biased results, as noted earlier. Therefore, it is of great interest to reanalyze the same data set without assuming certain information to be missing at random and then check the *sensitivity* of the results to the missing data mechanism. This is the main objective of this article.

When the missing data are nonignorable, we must model the missing data mechanism in order to obtain valid results. A common approach for the nonignorable missing data mechanism is to link the missing data indicator to the missing values and possibly other observed covariates via regression models such as a logistic regression model (e.g., Diggle & Kenward 1994; Ibrahim, Lipsitz & Chen 1999). Let $\mathbf{z}_i = (z_{i1}, \ldots, z_{ip})'$ be a $p \times 1$ vector of (time-independent) covariates for individual i (in this example, $p = 3$). We write $\mathbf{z}_i = (\mathbf{z}_{\text{mis},i}, \mathbf{z}_{\text{obs},i})$, where $\mathbf{z}_{\text{mis},i}$ corresponds to the missing components of the covariate vector $\mathbf{z}_i$ and $\mathbf{z}_{\text{obs},i}$ contains the observed components of $\mathbf{z}_i$. Let $\mathbf{r}_i = (r_{i1}, \ldots, r_{ip})'$ be a $p \times 1$ vector of missing data indicators such that $r_{ij} = 1$ if the jth covariate is missing for individual i and 0 otherwise, $j = 1, \ldots, p$.

To model the missing data mechanism, we may write the joint distribution of $\mathbf{r}_i$ as a product of one-dimensional conditional distributions and then use a logistic regression that links r_{ij} to the missing values $\mathbf{z}_{\text{mis},i}$ and the observed covariates $\mathbf{z}_{\text{obs},i}$ (e.g., Ibrahim, Lipsitz & Chen 1999). However, note that we should avoid building too large a model for the missing data mechanism since such models may become unidentifiable (Fitzmaurice, Laird & Zahner 1996). In the current example, since both the sample size and the intra-individual measurements are not large, we focus mainly on the following simple independent model

$$f(r_{i1}, r_{i2}, r_{i3} \mid \mathbf{z}_i, \phi) = \prod_{j=1}^{3} \{\mathrm{P}(r_{ij} = 1 \mid \mathbf{z}_i, \phi)\}^{r_{ij}} \{1 - \mathrm{P}(r_{ij} = 1 \mid \mathbf{z}_i, \phi)\}^{1-r_{ij}}, \quad (3)$$

$$\operatorname{logit}\{\mathrm{P}(r_{ij} = 1 \mid \mathbf{z}_i, \phi)\} = \phi_{0j} + \phi_j z_{ij}, \quad j = 1, 2, 3, \quad (4)$$

where $\phi = (\phi_{01}, \ldots, \phi_3)$. In models (3) and (4), we allow the missingness r_{ij} to possibly depend on z_{ij}, which may not be observed. Thus we allow the missing data mechanism to be nonignorable. We may also consider more elaborate models than (3) and (4), especially when the observed data are rich. For example, we may relax the independence assumption for the r_{ij} and let r_{ij} depend on r_{ik} ($k \neq j$), and we may include all three covariates and baseline response in (4) as explanatory variables. The parameters ϕ are treated as nuisance parameters and are often not of inferential interest. We often prefer parsimonious models for the missing data mechanism to reduce the number of nuisance parameters, as well as to avoid identifiability/estimability problems. Note that models (3) and (4) are not testable based on the observed data, so they should be viewed as a tool for sensitivity analysis.

Models (1)–(4) are highly complicated due to nonlinearity, random effects, and missing data. In the following sections, we consider general nonlinear mixed-effect models with nonignorably missing covariates and propose a Monte Carlo EM method to obtain the maximum likelihood estimates (MLE) of the model parameters.

3. A MONTE CARLO EXPECTATION MAXIMIZATION METHOD

3.1. The model and likelihood.

In this section, we consider general nonlinear mixed-effect models with nonignorably missing covariates. Suppose that there are N individuals, with individual i having n_i measurements

1708945x, 2004, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.2307/3315997 by Northeast Normal University, Wiley Online Library on [26/03/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

at times $t_{i1}, \ldots, t_{in_i}$. Let $\mathbf{y}_i = (y_{i1}, \ldots, y_{in_i})'$, where y_{ij} is the response value for individual i at time t_{ij}, $i = 1, \ldots, N$, $j = 1, \ldots, n_i$. Let f denote a generic density function and $y \mid x$ denote the conditional distribution of y given x. The observed data are $\{(\mathbf{y}_i, \mathbf{z}_{\text{obs},i}, \mathbf{r}_i), i = 1, \ldots, N\}$, where $\mathbf{z}_{\text{obs},i}$ and $\mathbf{r}_i$ are defined in the previous section.

We consider the following nonlinear mixed-effect model, which is written as a hierarchical two-stage model (Davidian & Giltinan 1995):

$$\begin{aligned} y_{ij} &= g(\mathbf{x}_{ij}, \boldsymbol{\beta}_i) + e_{ij}, \qquad \mathbf{e}_i \mid \boldsymbol{\beta}_i \sim \mathsf{N}(\mathbf{0}, \sigma^2 I), \\ \boldsymbol{\beta}_i &= \mathbf{h}(\mathbf{z}_i, \boldsymbol{\beta}) + B_i \mathbf{b}_i, \quad \mathbf{b}_i \text{ i.i.d.} \sim \mathsf{N}(\mathbf{0}, D), \quad j = 1, \ldots, n_i,\ i = 1, \ldots, N, \end{aligned}$$

where g is a nonlinear function, $\mathbf{x}_{ij}$ is a $v \times 1$ design vector of intra-individual variates such as time, $\mathbf{e}_i = (e_{i1}, \ldots, e_{in_i})'$, $\boldsymbol{\beta}_i = (\beta_{i1}, \ldots, \beta_{is})'$ is an $s \times 1$ vector of individual-specific regression parameters, $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_r)'$ is an $r \times 1$ vector of fixed effects, $\mathbf{h}$ is an s-dimensional vector-valued linear or nonlinear function, B_i is an incidence matrix of zeros and ones, $\mathbf{b}_i = (b_{i1}, \ldots, b_{is})'$ is the vector of random effects and is independent of $\mathbf{e}_i$, σ^2 is the unknown intra-individual variance, I is the identity matrix, and the $s \times s$ matrix D quantifies the interindividual covariance. We write $D = D(\boldsymbol{\eta})$, where $\boldsymbol{\eta}$ denotes the collection of all distinct parameters in D. The marginal density of the response $\mathbf{y}$ is given by

$$f(\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2, D) = \int f(\mathbf{y} \mid \mathbf{z}, \mathbf{b}, \boldsymbol{\beta}, \sigma^2) f(\mathbf{b} \mid D)\, d\mathbf{b},$$

which generally does not have a closed form expression. When missing covariates are present, likelihood inference becomes more complicated.

We consider a Monte Carlo EM algorithm for likelihood inference, following Ibrahim, Lipsitz & Chen (1999). We assume that the covariates $\mathbf{z}_i$ are independent of the random effects $\mathbf{b}_i$, and treat the unobservable random effects $\mathbf{b}_i$ as additional missing data. Thus, the complete data are $\{(\mathbf{y}_i, \mathbf{z}_i, \mathbf{b}_i, \mathbf{r}_i),\ i = 1, \ldots, N\}$. The complete-data log-likelihood can then be written as

$$\begin{aligned} \ell(\boldsymbol{\theta}) &\equiv \sum_{i=1}^{N} \ell(\boldsymbol{\theta}; \mathbf{y}_i, \mathbf{z}_i, \mathbf{b}_i, \mathbf{r}_i) \\ &= \sum_{i=1}^{N} \{\log f(\mathbf{y}_i \mid \mathbf{z}_i, \boldsymbol{\beta}, \sigma^2, \mathbf{b}_i) + \log f(\mathbf{z}_i \mid \boldsymbol{\alpha}) + \log f(\mathbf{b}_i \mid D) + \log f(\mathbf{r}_i \mid \mathbf{y}_i, \mathbf{z}_i, \boldsymbol{\phi})\}, \end{aligned}$$

where $\boldsymbol{\theta} = (\boldsymbol{\alpha}, \boldsymbol{\beta}, \sigma^2, \boldsymbol{\eta}, \boldsymbol{\phi})$ is the collection of all parameters, the $\boldsymbol{\alpha}$ are parameters for the covariates model $f(\mathbf{z}_i \mid \boldsymbol{\alpha})$, the $\boldsymbol{\phi}$ are parameters for the missing data model $f(\mathbf{r}_i \mid \mathbf{y}_i, \mathbf{z}_i, \boldsymbol{\phi})$, and both $\boldsymbol{\alpha}$ and $\boldsymbol{\phi}$ are treated as nuisance parameters. We also assume that the distribution of the missing data mechanism $\mathbf{r}_i$ does not depend on the random effects $\mathbf{b}_i$, and assume that the parameters $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, σ^2, $\boldsymbol{\eta}$, and $\boldsymbol{\phi}$ are all distinct.

To incorporate both categorical and continuous covariates in the covariate model $f(\mathbf{z}_i \mid \boldsymbol{\alpha})$, we may adapt the approach of Ibrahim, Lipsitz & Chen (1999), where the joint distribution of the covariates is modeled as a product of one-dimensional conditional distributions. We then apply common regression models such as logistic regressions and normal linear regressions to the conditional distributions (see Section 4 for details). We may consider similar approaches for the missing data model. To reduce the number of nuisance parameters in the covariate model and the missing data model, standard model selection methods such as the Akaike information criterion (AIC) or the Bayesian information criterion (BIC) and the likelihood ratio test can be used to select the most parsimonious model.

3.2. A Monte Carlo EM method.

For the Monte Carlo EM algorithm, in the E-step we use a Monte Carlo method to approximate the conditional expectation of the complete-data log-likelihood given the observed data and

1708945x, 2004, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.2307/3315997 by Northeast Normal University, Wiley Online Library on [26/03/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

current parameter estimates. The M-step is just like a complete-data maximization so standard complete-data methods can be used to update the parameter estimates. Thus we mainly focus on the E-step and discuss various sampling methods.

We denote the parameter estimates from the tth EM iteration by $\boldsymbol{\theta}^{(t)}$. Then, at the $(t+1)$st EM iteration, the E-step for individual i can be written as

$$\begin{aligned} Q_i(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(t)}) &= \mathrm{E}\left\{\ell(\boldsymbol{\theta}; \mathbf{y}_i, \mathbf{z}_i, \mathbf{r}_i, \mathbf{b}_i) \mid \text{ observed data}\right\} \\ &= \mathrm{E}\left\{\ell(\boldsymbol{\theta}; \mathbf{y}_i, \mathbf{z}_i, \mathbf{r}_i, \mathbf{b}_i) \mid \mathbf{y}_i, \mathbf{z}_{\mathrm{obs},i}, \mathbf{r}_i, \boldsymbol{\theta}^{(t)}\right\} \\ &= \iint \left\{\log f(\mathbf{y}_i \mid \mathbf{z}_i, \mathbf{b}_i, \boldsymbol{\beta}, \sigma^2) + \log f(\mathbf{z}_i \mid \boldsymbol{\alpha}) \right. \\ &\qquad \left. + \log f(\mathbf{b}_i \mid D) + \log f(\mathbf{r}_i \mid \mathbf{y}_i, \mathbf{z}_i, \boldsymbol{\phi})\right\} \\ &\qquad \times f(\mathbf{z}_{\mathrm{mis},i}, \mathbf{b}_i \mid \mathbf{z}_{\mathrm{obs},i}, \mathbf{y}_i, \mathbf{r}_i, \boldsymbol{\theta}^{(t)})\, d\mathbf{b}_i\, d\mathbf{z}_{\mathrm{mis},i} \equiv I_1 + I_2 + I_3 + I_4, \quad (5) \end{aligned}$$

which generally does not have a closed form expression. However, we may approximate the expectation in (5) using the Monte Carlo EM methods of Wei & Tanner (1990) and Ibrahim, Chen & Lipsitz (2001). Specifically, we use the Gibbs sampler (Gelfand & Smith 1990) to generate a sample from $\mathbf{z}_{\mathrm{mis},i}, \mathbf{b}_i \mid \mathbf{z}_{\mathrm{obs},i}, \mathbf{y}_i, \mathbf{r}_i, \boldsymbol{\theta}^{(t)}$ by sampling from the full conditionals $\mathbf{z}_{\mathrm{mis},i} \mid \mathbf{b}_i, \mathbf{z}_{\mathrm{obs},i}, \mathbf{y}_i, \mathbf{r}_i, \boldsymbol{\theta}^{(t)}$ and $\mathbf{b}_i \mid \mathbf{z}_{\mathrm{mis},i}, \mathbf{z}_{\mathrm{obs},i}, \mathbf{y}_i, \mathbf{r}_i, \boldsymbol{\theta}^{(t)}$. Note that

$$f(\mathbf{z}_{\mathrm{mis},i} \mid \mathbf{b}_i, \mathbf{z}_{\mathrm{obs},i}, \mathbf{y}_i, \mathbf{r}_i, \boldsymbol{\theta}^{(t)}) \propto f(\mathbf{z}_i \mid \boldsymbol{\alpha}^{(t)}) f(\mathbf{y}_i \mid \mathbf{z}_i, \mathbf{b}_i, \boldsymbol{\beta}^{(t)}, \sigma^{2(t)}) f(\mathbf{r}_i \mid \mathbf{y}_i, \mathbf{z}_i, \boldsymbol{\phi}^{(t)}), \quad (6)$$

$$f(\mathbf{b}_i \mid \mathbf{z}_{\mathrm{mis},i}, \mathbf{z}_{\mathrm{obs},i}, \mathbf{y}_i, \mathbf{r}_i, \boldsymbol{\theta}^{(t)}) \propto f(\mathbf{b}_i \mid D^{(t)}) f(\mathbf{y}_i \mid \mathbf{z}_i, \mathbf{b}_i, \boldsymbol{\beta}^{(t)}, \sigma^{2(t)}). \quad (7)$$

Monte Carlo samples from the full conditionals can be generated using rejection sampling or importance sampling methods. These methods are discussed in the next section.

Suppose that

$$(\tilde{\mathbf{z}}_{\mathrm{mis},i}^{(1)}, \tilde{\mathbf{b}}_i^{(1)}), \ldots, (\tilde{\mathbf{z}}_{\mathrm{mis},i}^{(m_i)}, \tilde{\mathbf{b}}_i^{(m_i)})$$

is a random sample of size m_i generated from $\mathbf{z}_{\mathrm{mis},i}, \mathbf{b}_i \mid \mathbf{z}_{\mathrm{obs},i}, \mathbf{y}_i, \boldsymbol{\theta}^{(t)}$ for individual i. Note that each $(\tilde{\mathbf{z}}_{\mathrm{mis},i}^{(k)}, \tilde{\mathbf{b}}_i^{(k)})$ depends on the EM iteration number, which is suppressed throughout. The E-step of the Monte Carlo EM algorithm at the $(t+1)$st iteration can be written as

$$\begin{aligned} Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(t)}) &= \sum_{i=1}^{N} Q_i(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(t)}) = \sum_{i=1}^{N} \left\{ \frac{1}{m_i} \sum_{j=1}^{m_i} \ell(\boldsymbol{\theta}; \mathbf{y}_i, \mathbf{z}_{\mathrm{obs},i}, \tilde{\mathbf{z}}_{\mathrm{mis},i}^{(j)}, \tilde{\mathbf{b}}_i^{(j)}) \right\} \\ &= \sum_{i=1}^{N} \sum_{j=1}^{m_i} \frac{1}{m_i} \log f(\mathbf{y}_i \mid \mathbf{z}_{\mathrm{obs},i}, \tilde{\mathbf{z}}_{\mathrm{mis},i}^{(j)}, \boldsymbol{\beta}, \sigma^2, \tilde{\mathbf{b}}_i^{(j)}) \\ &\quad + \sum_{i=1}^{N} \sum_{j=1}^{m_i} \frac{1}{m_i} \log f(\mathbf{z}_{\mathrm{obs},i}, \tilde{\mathbf{z}}_{\mathrm{mis},i}^{(j)} \mid \boldsymbol{\alpha}) \\ &\quad + \sum_{i=1}^{N} \sum_{j=1}^{m_i} \frac{1}{m_i} \log f(\tilde{\mathbf{b}}_i^{(j)} \mid D) + \sum_{i=1}^{N} \sum_{j=1}^{m_i} \frac{1}{m_i} \log f(\mathbf{r}_i \mid \mathbf{y}_i, \mathbf{z}_{\mathrm{obs},i}, \tilde{\mathbf{z}}_{\mathrm{mis},i}^{(j)}, \boldsymbol{\phi}) \\ &= Q^{(1)}(\boldsymbol{\beta}, \sigma^2 \mid \boldsymbol{\theta}^{(t)}) + Q^{(2)}(\boldsymbol{\alpha} \mid \boldsymbol{\theta}^{(t)}) + Q^{(3)}(D \mid \boldsymbol{\theta}^{(t)}) + Q^{(4)}(\boldsymbol{\phi} \mid \boldsymbol{\theta}^{(t)}). \end{aligned}$$

The M-step of the Monte Carlo EM algorithm is then like a complete-data maximization, where the missing data are substituted by Monte Carlo values. Since the parameters in $Q^{(1)}$, $Q^{(2)}$, $Q^{(3)}$ and $Q^{(4)}$ are distinct, we may maximize each $Q^{(k)}$ separately using standard procedures for the corresponding complete-data models, such as nonlinear and linear normal regression models and logistic regression models.

1708945x, 2004, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.2307/3315997 by Northeast Normal University, Wiley Online Library on [26/03/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

To obtain the variance-covariance of θ, we may use the formula of Louis (1982). Let $\widehat{\theta}$ be the estimate of θ at convergence. The observed information matrix can be estimated by

$$I(\widehat{\theta}) = -\ddot{Q}(\widehat{\theta} \mid \widehat{\theta}) - \left\{ \sum_{i=1}^{N} \sum_{j=1}^{m_i} \frac{1}{m_i} S_{ij}(\widehat{\theta}) S'_{ij}(\widehat{\theta}) - \sum_{i=1}^{N} \dot{Q}_i(\widehat{\theta} \mid \widehat{\theta}) \dot{Q}'_i(\widehat{\theta} \mid \widehat{\theta}) \right\},$$

where

$$S_{ij}(\widehat{\theta}) = \frac{\partial}{\partial \theta} \ell(\theta; \mathbf{y}_i, \mathbf{z}_{\text{obs},i}, \widetilde{\mathbf{z}}^{(j)}_{\text{mis},i}, \widetilde{\mathbf{b}}^{(j)}_i, \mathbf{r}_i) \mid_{\theta=\hat{\theta}},$$

$$\dot{Q}_i(\widehat{\theta} \mid \widehat{\theta}) = \sum_{j=1}^{m_i} \frac{1}{m_i} S_{ij}(\widehat{\theta}), \quad \ddot{Q}(\widehat{\theta} \mid \widehat{\theta}) = \sum_{i=1}^{N} \sum_{j=1}^{m_i} \frac{1}{m_i} \frac{\partial}{\partial \theta} S_{ij}(\theta) \mid_{\theta=\hat{\theta}}.$$

Thus, the asymptotic covariance matrix of $\widehat{\theta}$ is $I^{-1}(\widehat{\theta})$, and the asymptotic covariance matrix of β is the upper $r \times r$ block of $I^{-1}(\widehat{\theta})$.

When implementing the Monte Carlo E-step, it is necessary to choose the numbers of Monte Carlo samples m_i. It is obvious that larger values of m_i will result in more exact but slower computations. A common strategy is to increase m_i as the number of EM iterations increase (Booth & Hobert 1999). Note that, for Monte Carlo EM algorithms, the incomplete-data log-likelihood is not guaranteed to increase at each iteration due to Monte Carlo error at the E-step. Under suitable regularity conditions, however, the Monte Carlo EM algorithms still converge to the maximum likelihood estimate (Chan & Ledolter 1995).

Instead of choosing m_i in a subjective manner, Booth & Hobert (1999) proposed an automated method which can be extended to our case as follows. Let

$$Q^{(1)}(\theta \mid \hat{\theta}^{(t)}) = \frac{\partial}{\partial \theta} Q(\theta \mid \hat{\theta}^{(t)}), \quad Q^{(2)}(\theta \mid \hat{\theta}^{(t)}) = \frac{\partial^2}{\partial \theta \partial \theta'} Q(\theta \mid \hat{\theta}^{(t)}),$$

and let $\theta^{*(t+1)}$ be the solution of $Q^{(1)}(\theta \mid \hat{\theta}^{(t)}) = 0$. When the simulated samples are independent, by the central limit theorem, the conditional distribution of $\theta^{(t+1)} \mid \theta^{(t)}$ is approximately normal with mean $\theta^{*(t+1)}$, and a variance that can be estimated by

$$\widehat{\text{var}}(\theta^{(t+1)} \mid \theta^{(t)}) = Q^{(2)}(\theta^{*(t+1)} \mid \hat{\theta}^{(t)})^{-1} \widehat{\text{var}} \{ Q_i^{(1)}(\theta^{*(t+1)} \mid \hat{\theta}^{(t)}) \} Q^{(2)}(\theta^{*(t+1)} \mid \hat{\theta}^{(t)})^{-1},$$

where

$$\widehat{\text{var}} \{ Q_i^{(1)}(\theta^{*(t+1)} \mid \hat{\theta}^{(t)}) \} = \frac{1}{m_i} \sum_{j=1}^{m_i} \left\{ \tilde{w}_{tj} \frac{\partial}{\partial \theta} \log f(\mathbf{y}, \mathbf{z}_{\text{obs}}, \tilde{\mathbf{z}}^{(j)}_{\text{mis}}, \tilde{\mathbf{b}}^{(j)}_i; \theta^{(t+1)}) \right\} \times \left\{ \tilde{w}_{tj} \frac{\partial}{\partial \theta} \log f(\mathbf{y}, \mathbf{z}_{\text{obs}}, \tilde{\mathbf{z}}^{(j)}_{\text{mis}}, \tilde{\mathbf{b}}^{(j)}_i; \theta^{(t+1)}) \right\}',$$

$\tilde{\mathbf{z}}^{(j)}_{\text{mis},i}$ and $\tilde{\mathbf{b}}^{(j)}_i$ are simulated samples, and $\tilde{w}_{tj}$ are the importance weights when the importance sampling method is used and are all set equal to one when rejection sampling methods are used. After the $(t+1)$st iteration, we may then construct an approximate $100(1-\alpha)\%$ confidence ellipsoid for $\theta^{*(t+1)}$ based on the above normal approximation. The EM step is swamped by Monte Carlo error if the previous value $\theta^{(t)}$ lies in the confidence ellipsoid, and in that case we need to increase m_i (e.g., set m_i to be $m_i + m_i/k$ for some positive constant k).

3.3. Generation of Monte Carlo samples.

We now discuss several methods for sampling the full conditionals in (6) and (7) in the E-step. If the appropriate densities are log-concave, the Gibbs sampler along with the adaptive rejection algorithm of Gilks & Wild (1992) may be used. Specifically, if $f(\mathbf{z}_i \mid \alpha^{(t)})$,

1708945x, 2004, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.2307/3315997 by Northeast Normal University, Wiley Online Library on [26/03/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

$f(\mathbf{y}_i \mid \mathbf{z}_i, \mathbf{b}_i, \boldsymbol{\beta}^{(t)}, \sigma^{2(t)})$, and $f(\mathbf{r}_i \mid \mathbf{y}_i, \mathbf{z}_i, \boldsymbol{\theta}^{(t)})$ are log-concave in each component of $\mathbf{z}_i$ and $f(\mathbf{b}_i \mid D^{(t)})$ and $f(\mathbf{y}_i \mid \mathbf{z}_i, \mathbf{b}_i, \boldsymbol{\beta}^{(t)}, \sigma^{2(t)})$ are log-concave in each component of $\mathbf{b}_i$, we can use the Gibbs sampler along with the adaptive rejection algorithm to sample from the full conditionals $f(\mathbf{z}_{\text{mis},i} \mid \mathbf{b}_i, \mathbf{z}_{\text{obs},i}, \mathbf{y}_i, \mathbf{r}_i, \boldsymbol{\theta}^{(t)})$ and $f(\mathbf{b}_i \mid \mathbf{z}_{\text{mis},i}, \mathbf{z}_{\text{obs},i}, \mathbf{y}_i, \mathbf{r}_i, \boldsymbol{\theta}^{(t)})$, as in Ibrahim, Chen & Lipsitz (2001).

Sometimes, however, the log-concavity may not be satisfied. In such cases, we may consider the method of Evans & Swartz (1998), which extends the algorithm of Gilks and Wild (1992) to more general cases. The adaptive rejection method of Evans and Swartz does not require the log-concave restriction on the densities. Alternatively, multivariate rejection sampling (e.g., Geweke 1996, §3.2) may be used to obtain the desirable samples, as in Booth & Hobert (1999). For example, consider sampling from $f(\mathbf{b}_i \mid \mathbf{z}_{\text{mis},i}, \mathbf{z}_{\text{obs},i}, \mathbf{y}_i, \mathbf{r}_i, \boldsymbol{\theta}^{(t)})$. Let $f^*(\mathbf{b}_i) = f(\mathbf{y}_i \mid \mathbf{z}_i, \mathbf{b}_i, \boldsymbol{\beta}^{(t)}, \sigma^{2(t)})$ and $\tau = \sup_{\mathbf{u}}\{f^*(\mathbf{u})\}$. A random sample from $f(\mathbf{b}_i \mid \mathbf{z}_{\text{mis},i}, \mathbf{z}_{\text{obs},i}, \mathbf{y}_i, \mathbf{r}_i, \boldsymbol{\theta}^{(t)})$ can be obtained as follows.

Step 1: Sample $\mathbf{b}_i^*$ from $f(\mathbf{b}_i \mid D^{(t)})$, and independently, sample w from the uniform distribution on the interval $(0, 1)$.

Step 2: If $w \le f^*(\mathbf{b}_i^*)/\tau$, then accept $\mathbf{b}_i^*$, otherwise, go to Step 1.

Samples from $f(\mathbf{z}_{\text{mis},i} \mid \mathbf{b}_i, \mathbf{z}_{\text{obs},i}, \mathbf{y}_i, \mathbf{r}_i, \boldsymbol{\theta}^{(t)})$ can be obtained in a similar way.

When the dimensions of $\mathbf{z}_{\text{mis},i}$ or $\mathbf{b}_i$ are large, the rejection sampling methods may be slow. In this case, following Booth & Hobert (1999), we may use importance sampling methods where the importance function can be chosen to be a multivariate Student t density or a multivariate normal density whose mean and variance match the mode and curvature of $f(\mathbf{z}_{\text{mis},i}, \mathbf{b}_i \mid \mathbf{z}_{\text{obs},i}, \mathbf{y}_i, \mathbf{r}_i, \boldsymbol{\theta}^{(t)})$. Specifically, we write

$$f(\mathbf{z}_{\text{mis},i}, \mathbf{b}_i \mid \mathbf{z}_{\text{obs},i}, \mathbf{y}_i, \mathbf{r}_i, \boldsymbol{\theta}^{(t)}) = c \exp\{q(\mathbf{z}_{\text{mis},i}, \mathbf{b}_i)\},$$

where c is the unknown normalizing constant. Let $q^{(1)}(\mathbf{z}_{\text{mis},i}, \mathbf{b}_i)$ and $q^{(2)}(\mathbf{z}_{\text{mis},i}, \mathbf{b}_i)$ be the first and second derivatives of $q(\mathbf{z}_{\text{mis},i}, \mathbf{b}_i)$, respectively. Let $(\widetilde{\mathbf{z}}_{\text{mis},i}, \widetilde{\mathbf{b}}_i)$ be the solution of $q^{(1)}(\mathbf{z}_{\text{mis},i}, \mathbf{b}_i) = 0$ which maximizes $q(\mathbf{z}_{\text{mis},i}, \mathbf{b}_i)$. Then, the Laplace approximations of the mean and variance of $f(\mathbf{z}_{\text{mis},i}, \mathbf{b}_i \mid \mathbf{z}_{\text{obs},i}, \mathbf{y}_i, \mathbf{r}_i, \boldsymbol{\theta}^{(t)})$ are $(\widetilde{\mathbf{z}}_{\text{mis},i}, \widetilde{\mathbf{b}}_i)$ and $-\{q^{(2)}(\widetilde{\mathbf{z}}_{\text{mis},i}, \widetilde{\mathbf{b}}_i)\}^{-1}$, respectively (Booth & Hobert 1999).

Suppose that

$$(\widetilde{\mathbf{z}}_{\text{mis},i}^{*(1)}, \widetilde{\mathbf{b}}_i^{*(1)}), \ldots, (\widetilde{\mathbf{z}}_{\text{mis},i}^{*(m_i)}, \widetilde{\mathbf{b}}_i^{*(m_i)})$$

is a random sample of size m_i generated from the importance function $h^*(\mathbf{z}_{\text{mis},i}, \mathbf{b}_i)$, which is assumed to have the same support as $f(\mathbf{z}_{\text{mis},i}, \mathbf{b}_i \mid \mathbf{z}_{\text{obs},i}, \mathbf{y}_i, \mathbf{r}_i, \boldsymbol{\theta}^{(t)})$. Then we have

$$Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(t)}) \approx \sum_{i=1}^{N} \left\{ \frac{1}{m_i} \sum_{j=1}^{m_i} w_{ij}^{(t)} \ell(\boldsymbol{\theta}; \mathbf{y}_i, \mathbf{z}_{\text{obs},i}, \widetilde{\mathbf{z}}_{\text{mis},i}^{*(j)}, \widetilde{\mathbf{b}}_i^{*(j)}) \right\},$$

where $w_{ij}^{(t)} = f(\widetilde{\mathbf{z}}_{\text{mis},i}^{*(j)}, \widetilde{\mathbf{b}}_i^{*(j)} \mid \mathbf{z}_{\text{obs},i}, \mathbf{y}_i, \boldsymbol{\theta}^{(t)})/h^*(\widetilde{\mathbf{z}}_{\text{mis},i}^{*(j)}, \widetilde{\mathbf{b}}_i^{*(j)})$ are importance weights.

We may also consider other sampling methods such as those based on Metropolis–Hastings or Markov chains (e.g., McCulloch 1997). However, these methods often generate *dependent* samples which may not lead to straightforward assessment of Monte Carlo error (Booth & Hobert 1999). The rejection and importance sampling methods on the other hand, generate *independent* samples which can be used to assess Monte Carlo error in a straightforward way, as in Booth & Hobert (1999). Among the rejection and importance sampling methods discussed in this section, based on our limited experience, the rejection sampling methods (especially the

1708945x, 2004, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.2307/3315997 by Northeast Normal University, Wiley Online Library on [26/03/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

adaptive methods) may be more efficient when the dimension of the random effects and the sample sizes are small, while the importance sampling method may be more efficient when the sample sizes are large: in this case the importance distribution may closely resemble the true conditional distribution. Further research is needed to compare these methods thoroughly.

4. ANALYSIS

In this section, we analyze the data set described in Section 2. Our objective is to check the sensitivity of the results to the missing data mechanism. We consider estimating the population parameters

$$\beta = (\log(P_{10}), a_{11}, \lambda_{10}, a_{21}, a_{22}, \log(P_{20}), a_{31}, \lambda_{20}, a_{41})$$

using three methods: the complete-case method, the EM method assuming an ignorable missing mechanism, and the EM method assuming a nonignorable missing mechanism. The complete-case method is based on the approximate method of Lindstrom & Bates (1990). To avoid very small (large) estimates, which may be unstable, we standardized the covariate values and rescaled the original time t (in days) so that the new time scale is between 0 and 1. This data set was previously analyzed under the ignorable missing data assumption, and models (1) and (2) were shown to fit well, based on the complete data (Wu & Ding 1999; Wu & Wu 2001). Thus we do not repeat the model diagnostics here.

Because all three covariates contain missing data, we must specify their distribution for the analysis. We model the joint distribution of the three covariates as a product of three one-dimensional conditional distributions:

$$f(z_{i1}, z_{i2}, z_{i3} \mid \boldsymbol{\alpha}) = f(z_{i3} \mid z_{i1}, z_{i2}, \boldsymbol{\alpha}_3) f(z_{i2} \mid z_{i1}, \boldsymbol{\alpha}_2) f(z_{i1} \mid \boldsymbol{\alpha}_1).$$

Since all three covariates are continuous and each marginal distribution is approximately normal, we assume a linear normal model for each conditional distribution. That is, we assume

$$z_{i3} \mid z_{i1}, z_{i2}, \boldsymbol{\alpha}_3 \sim \mathsf{N}(\alpha_{30} + \alpha_{31} z_{i1} + \alpha_{32} z_{i2}, \alpha_{33}), \quad z_{i2} \mid z_{i1}, \boldsymbol{\alpha}_2 \sim \mathsf{N}(\alpha_{20} + \alpha_{21} z_{i1}, \alpha_{22}),$$

and

$$z_{i1} \mid \boldsymbol{\alpha}_1 \sim \mathsf{N}(\alpha_{10}, \alpha_{11}),$$

where $\boldsymbol{\alpha}_3 = (\alpha_{30}, \alpha_{31}, \alpha_{32}, \alpha_{33})$, $\boldsymbol{\alpha}_2 = (\alpha_{20}, \alpha_{21}, \alpha_{22})$, and $\boldsymbol{\alpha}_1 = (\alpha_{10}, \alpha_{11})$.

One advantage of this approach is that we may use standard variable selection methods (e.g., AIC, BIC, and the likelihood ratio test) for the regression models to reduce the number of nuisance parameters, as will be shown later in this section. Another advantage is that it allows the missing covariates to be a mixture of continuous and categorical variables. For the missing data mechanism, we use the independent logistic models (3) and (4) as discussed in Section 2. The EM method with ignorable missing mechanism is obtained by setting $\phi_1 = \phi_2 = \phi_3 = 0$. We take 500 Monte Carlo samples within each EM iteration. We assess the convergence of the Markov chains (Gibbs sampler) by examining time-series plots and sample autocorrelations function, and conservatively choose 500 iterations for burn-in. The starting values for the EM algorithms are taken to be the estimates based on the complete case method.

Table 1 gives the results of the analysis. We see that the estimates based on the EM methods with ignorable and nonignorable missing data mechanisms are quite different in some cases. For example, the association of TNF and P_{1i} is not significant at the 5% level under the nonignorable model but is significant under the ignorable model, and the association of TNF and λ_{1i} may be overestimated (biased) under the ignorable model. The standard errors under the nonignorable model may be much larger in some cases, probably due in part to the incorporation of the missing data model which introduces additional nuisance parameters. Based on our limited experience, this effect of additional nuisance parameters can be large for nonlinear models with relatively sparse observed data such as ours. The complete-case method also produces somewhat

1708945x, 2004, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.2307/3315997 by Northeast Normal University, Wiley Online Library on [26/03/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

different results—possibly biased estimates and large standard errors. It is well known that the complete-case method is not only inefficient due to discarding incomplete observations but may also give biased results if the missing data are not missing at random (even when the missing data are ignorable).

TABLE 1: Estimates and standard errors for the viral dynamic parameters.

Parameter	Nonignorable model		Ignorable model		Complete case	
$\log(P_{10})$	12.56	0.53	12.26	0.21	12.65	0.30
a_{11}	0.75	0.50	0.81	0.25	0.54	0.26
λ_{10}	14.97	2.79	13.06	0.77	12.28	0.98
a_{21}	0.26	1.40	1.60	0.93	0.14	1.11
a_{22}	2.68	1.52	2.00	0.61	1.24	0.65
$\log(P_{20})$	8.74	0.61	7.81	0.24	8.38	0.38
a_{31}	0.52	0.16	0.58	0.08	0.43	0.18
λ_{20}	1.83	0.79	0.93	0.28	1.50	0.40
a_{41}	−0.16	0.11	−0.18	0.10	−0.28	0.18

We also checked the sensitivity of our results to (the choices of) the covariates model and the missing data model. For the covariate model, we performed variable selections on each regression model using the stepwise method and obtained a reduced covariate model which has two fewer (nuisance) parameters ($\alpha_{21} = \alpha_{31} = 0$). We then refitted the nonlinear mixed-effect model with the reduced covariate model. For the missing data model, we included the baseline viral loads and all three covariates in the logistic models (4). All these analyses produced similar parameter estimates. Thus the results in Table 1 may be robust to the specifications of the covariate model and the missing data model.

The above computations were implemented by the Gibbs sampler along with the adaptive rejection method since the density functions are log-concave in the appropriate parameters. The models were fitted using computer code written in FORTRAN and S-PLUS. The code and the data set are available upon request from the author. The EM methods converged in several hours on a SUN Sparc workstation. We also compared the adaptive rejection method with the multivariate rejection method and found that the multivariate rejection method converged more slowly for this example. This is probably because the dimension of the random effects is not small here, so the acceptance rate is very low for the multivariate rejection algorithm. To check whether this is the case, we fitted the nonlinear mixed-effect model with the same stage-one model (1), but for the stage-two model we only included one random effect for λ_{1i} (i.e., we set $b_{1i} = b_{3i} = b_{4i} = 0$). Thus the dimension of the random effects is reduced to one from four. In this case, we found that the multivariate rejection method converges quite quickly. We also found that the importance sampling method was slow for this example. Thus, for this example, the adaptive rejection method appeared to perform the best. We are currently conducting a thorough investigation in order to further compare these sampling methods under various situations.

In conclusion, the proposed nonignorable model may be used as a tool for sensitivity analysis regarding the missing data mechanism. For the data set considered here, the estimates of the viral dynamic parameters appear to be sensitive to the missing data mechanism. Obtaining reliable estimates of the viral dynamic parameters is important for clinical decisions. Our results suggest that, in the presence of missing covariates, the missing data mechanism may need to be incorporated into the analyses. Therefore, previous results assuming an ignorable missing data mechanism may not be reliable and further investigation may be needed for this study. Finally, note that the assumption for the missing data mechanism cannot be tested based on the observed

1708945x, 2004, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.2307/3315997 by Northeast Normal University, Wiley Online Library on [26/03/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

data and the foregoing analyses based on the nonignorable models may be model-dependent, so our results can only provide rough guidance regarding the missing data mechanism and suggest the need for further investigation.

5. DISCUSSION

We have considered a Monte Carlo EM method for nonlinear mixed-effect models with nonignorably missing covariates. Sometimes the method may suffer from convergence problems such as slow or nonconvergence, especially when the dimension of random effects is large or the intra-individual data are sparse. In such cases, we may consider approximate methods such as those based on Lindstrom & Bates (1990). Approximate methods for nonlinear mixed-effect models with nonignorably missing covariates are currently under investigation.

The nonignorable models described in this article should be viewed as a tool for sensitivity analysis regarding the missing data mechanism since such models are not testable based on the observed data. The parameters in the covariate and the missing data models are viewed as nuisance parameters. We should try to reduce the number of nuisance parameters to avoid identifiability problems and to improve the efficiency of the parameter estimates which are of main interest for inference. This can be achieved by building parsimonious models for the covariate and missing data models using standard variable selection methods for regression models. Finally, the method may be extended to nonlinear mixed-effect models with missing time-dependent covariates. In this case, a repeated measurement model may be needed for the incompletely observed covariates.

ACKNOWLEDGEMENTS

I am grateful to Professors Ming-Hui Chen, Joseph G. Ibrahim, Xiao-Li Meng, and Hulin Wu for help in many ways, and to the Editor, the Associate Editor, and two referees for very constructive suggestions that greatly improved the article. Partial funding in support of this work was provided by the Natural Sciences and Engineering Research Council of Canada grant no 22R80742.

REFERENCES

J. G. Booth & J. P. Hobert (1999). Maximizing generalized linear mixed models likelihoods with an automated Monte Carlo EM algorithm. *Journal of the Royal Statistical Society Series B*, 61, 265–285.

K. S. Chan & J. Ledolter (1995). Monte Carlo EM estimation for time series models involving counts. *Journal of the American Statistical Association*, 90, 242–252.

H. Y. Chen & R. J. A. Little (1999). Proportional hazards regression with missing covariates. *Journal of the American Statistical Association*, 94, 896–908.

M. Davidian & D. M. Giltinan (1995). *Nonlinear Models for Repeated Measurements Data.* Chapman & Hall, London.

P. Diggle & M. G. Kenward (1994). Informative drop-out in longitudinal data analysis (with discussion). *Applied Statistics: Journal of the Royal Statistical Society Series C*, 43, 49–93.

M. J. Evans & T. Swartz (1998). Random variable generation using concavity properties of transformed densities. *Journal of Computational and Graphical Statistics*, 7, 514–528.

G. M. Fitzmaurice, N. M. Laird & G. E. P. Zahner (1996). Multivariate logistic models for incomplete binary responses. *Journal of the American Statistical Association*, 91, 99–108.

A. E. Gelfand & A. F. M. Smith (1990). Sampling-based approaches to calculating marginal densities. *Journal of the American Statistical Association*, 85, 398–409.

J. Geweke (1996). Monte-Carlo simulation and numerical integration. Chapter 15 in *Handbook of Computational Economics* (H. M. Amman, D. A. Kendrick & J. Rust, eds.), North-Holland, Amsterdam, pp. 731–797.

W. R. Gilks & P. Wild (1992). Adaptive rejection sampling for Gibbs sampling. *Applied Statistics: Journal of the Royal Statistical Society Series C*, 41, 337–348.

1708945x, 2004, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.2307/3315997 by Northeast Normal University, Wiley Online Library on [26/03/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

D. D. Ho, A. V. Neumann, A. S. Perelson, W. Chen, J. M. Leonard & M. Markowitz (1995). Rapid turnover of plasma virions and CD4 lymphocytes in HIV-1 infection. *Nature*, 373, 123–126.

J. G. Ibrahim, M. H. Chen & S. R. Lipsitz (2001). Missing responses in generalized linear mixed models when the missing data mechanism is nonignorable. *Biometrika*, 88, 551–564.

J. G. Ibrahim, S. R. Lipsitz & M. H. Chen (1999). Missing covariates in generalized linear models when the missing data mechanism is nonignorable. *Journal of the Royal Statistical Society Series B*, 61, 173–190.

M. J. Lindstrom & D. M. Bates (1990). Nonlinear mixed effects models for repeated measures data. *Biometrics*, 46, 673–687.

R. J. A. Little (1992). Regression with missing X's: a review. *Journal of the American Statistical Association*, 87, 1227–1237.

R. J. A. Little & D. B. Rubin (1987). *Statistical Analysis with Missing Data.* Wiley, New York.

T. A. Louis (1982). Finding the observed information matrix when using the EM algorithm. *Journal of the Royal Statistical Society Series B*, 44, 226–233.

C. E. McCulloch (1997). Maximum likelihood algorithms for generalized linear mixed models. *Journal of the American Statistical Association*, 92, 162–170.

A. S. Perelson, A. V. Neumann, M. Markowitz, J. M. Leonard & D. D. Ho (1996). HIV-1 dynamics in vivo: virion clearance rate, infected cell life-span, and viral generation time. *Science*, 271, 1582–1586.

E. F. Vonesh & V. M. Chinchilli (1997). *Linear and Nonlinear Models for the Analysis of Repeated Measurements*, Marcel Dekker, New York.

G. C. Wei & M. A. Tanner (1990). A Monte Carlo implementation of the EM algorithm and the poor man's data augmentation algorithm. *Journal of the American Statistical Association*, 85, 699–704.

H. Wu & A. Ding (1999). Population HIV-1 dynamics in vivo: applicable models and inferential tools for virological data from AIDS clinical trials. *Biometrics*, 55, 410–418.

H. Wu & L. Wu (2001). A multiple imputation method for missing covariates in nonlinear mixed-effect models, with application to HIV dynamics. *Statistics in Medicine*, 20, 1755–1769.

Received 5 November 2002
Accepted 2 December 2003

Lang WU: lang@stat.ubc.ca
Department of Statistics
The University of British Columbia
Vancouver, British Columbia, Canada V6T 1Z2

1708945x, 2004, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.2307/3315997 by Northeast Normal University, Wiley Online Library on [26/03/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License